

SANTEE COOPER, SC, USA

WMO: 723107

Lat: 33.587N

Lon: 80.209W

Elev: 102

StdP: 14.64

Time Zone: -5.00 (NAE)

Period: 09-19

WBAN: 00371

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	26.5	28.2	1.0	5.8	43.1	9.1	8.8	41.4	17.9	50.9	16.2	51.5	1.8	0	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.5	95.4	76.8	92.9	76.1	90.7	75.5	82.1	87.4	80.9	85.6	79.3	84.4	6.5	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
81.2	162.6	84.8	79.4	152.9	82.8	78.5	148.7	82.1	45.9	89.3	44.4	84.5	42.9	84.3	85.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.9	13.2	11.3		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	64.6	46.6	51.8	56.1	64.1	72.2	78.6	81.2	79.3	75.1	65.1	53.9	50.2
	DBStd	14.10	10.06	9.03	9.14	7.83	6.21	4.39	4.36	4.27	4.62	7.45	7.87	9.23
	HDD50	475	188	79	44	4	0	0	0	0	0	2	43	115
	HDD65	2277	574	377	299	110	14	0	0	1	91	344	466	
	CDD50	5797	82	130	234	427	688	858	967	908	753	469	159	122
	CDD65	2127	4	7	25	83	238	408	502	444	305	94	10	7
	CDH74	19820	19	64	234	742	2111	4001	5313	4107	2436	712	61	20
	CDH80	7931	0	4	44	178	789	1698	2475	1724	877	139	3	0
(14) Wind	WSAvg	4.0	3.8	4.3	4.6	5.1	4.7	4.3	4.3	3.2	3.3	3.3	3.5	3.6
(15) Precipitation	PrecAvg	51.1	3.7	3.8	3.6	3.2	3.7	5.7	5.9	6.8	4.4	3.6	2.9	3.5
	PrecMax	65.5	7.5	8.9	6.1	6.8	7.5	13.6	11.5	10.7	8.4	10.6	6.8	7.2
	PrecMin	38.6	1.0	1.3	0.6	0.9	1.3	2.3	2.7	4.1	0.6	0.0	0.2	1.2
	PrecStd	7.4	2.0	1.8	1.6	1.6	1.5	2.8	2.2	2.1	2.2	2.5	1.8	1.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	77.0	80.7	84.4	88.0	95.3	99.0	99.5	97.5	91.4	88.0	80.7	77.1
		MCWB	62.6	61.6	64.8	71.0	72.9	75.5	77.9	76.8	74.2	72.7	67.3	66.1
	2%	DB	70.4	75.4	80.9	84.5	90.6	94.6	96.6	93.1	90.1	82.5	74.9	72.5
		MCWB	60.2	60.3	63.2	68.8	70.5	75.7	77.1	76.9	74.1	70.0	63.7	64.1
	5%	DB	66.4	72.1	76.7	81.6	88.0	90.8	93.1	90.7	88.1	81.0	72.1	69.8
		MCWB	57.2	59.0	61.8	66.4	70.5	74.1	77.4	76.8	73.7	69.2	62.7	64.5
	10%	DB	63.2	68.1	72.2	78.9	85.5	89.5	90.7	88.5	85.9	78.6	68.4	65.7
		MCWB	56.6	57.8	58.8	65.1	70.0	73.7	77.2	75.8	73.0	67.9	61.8	61.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	66.2	67.0	69.0	75.7	77.0	82.8	84.2	83.6	80.7	78.0	71.2	70.3
		MCDB	71.6	74.0	73.8	83.0	84.6	91.0	91.9	89.1	83.3	80.4	76.0	73.0
	2%	WB	63.3	64.3	67.1	73.1	75.3	80.0	82.4	81.9	78.9	74.6	68.7	68.4
		MCDB	66.6	70.3	74.3	79.4	84.3	84.5	88.1	86.5	84.2	80.3	71.6	70.5
	5%	WB	61.2	62.2	65.0	70.9	73.9	78.2	81.2	80.6	77.6	72.6	65.8	65.1
		MCDB	65.5	68.9	73.1	77.5	82.5	83.6	86.0	85.1	83.9	77.7	69.8	67.8
	10%	WB	57.7	59.5	62.7	68.7	72.5	76.8	79.7	78.7	75.7	70.8	63.0	62.2
		MCDB	62.8	65.6	69.2	75.4	80.9	83.8	85.2	83.9	82.8	75.8	66.3	65.2
(35) Mean Daily Temperature Range	5% DB	MDBR	20.4	22.2	23.2	23.2	20.7	19.2	18.5	18.0	19.1	21.6	22.6	19.6
		MCDBR	23.6	25.9	26.1	24.9	23.3	21.0	20.2	20.1	21.7	22.6	25.2	22.5
		MCWBR	15.5	16.6	12.0	11.6	11.0	8.7	7.3	7.6	9.9	12.8	16.4	16.2
	5% WB	MCDBR	20.0	22.8	22.5	20.5	20.1	18.2	17.3	17.2	19.8	17.8	19.6	18.1
		MCWBR	15.8	17.0	12.3	11.4	9.7	9.5	8.8	9.1	10.9	11.8	14.4	14.7
(40) Clear-Sky Solar Irradiance	taub		0.321	0.329	0.364	0.408	0.475	0.507	0.551	0.520	0.436	0.392	0.344	0.319
	taud		2.514	2.487	2.413	2.279	2.139	2.095	1.993	2.070	2.296	2.390	2.485	2.554
	Ebn at Noon		284	293	289	279	261	251	239	245	262	266	272	278
	Edh at Noon		27	30	35	42	49	51	56	51	39	33	27	24
(44) All-Sky Solar Radiation	RadAvg		892	1092	1441	1823	1965	2028	1947	1754	1508	1275	983	790
	RadStd		48	111	115	124	148	141	125	91	129	151	75	68

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (59 neighbors)	+1.01	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+346	+243

Nomenclature: See separate page